some other purposes. More projects are being developed for heavy traffic areas for testing.

There are many ongoing efforts around the world to create smart highways with specific focus on solar roads and smart traffic for AVs. Smart highway sensors and software programs would allow real-time feedback for safety features to prevent or mitigate accidents.

Robotic nurses in a hospital in Bangkok (Credit: www.straitstimes.com)

Solar road projects are also on the way in countries like the US, Georgia, France and China. Currently, China has the largest stretch of solar road with an area of 5875 square-metres. The world's first solar highway in Shandong Province was officially put into use on December 28, 2017.

Solar roads could be used not only to power electric vehicles (EVs) without batteries but also to charge batteries connected to vehicles. Other potential uses for solar roads include powering outdoor vending machines, safety systems and road lighting. Safety systems use SmartLife poles that exchange data between poles to indicate oncoming traffic. China has been at the fore front in building robust mechanisms to implement such technologies on a large-scale, enabling smart highways and roads.

Drones in logistics

Companies including Amazon are paving the way for drones in logistics. Drones are not only used for outdoor deliveries but for running a warehouse, too. Combined with good logistics software, drones can drastically improve efficiency of entire logistics operations.

Use of drones in public air traffic is still a problem in most countries. Some companies have already started using drones within their own four walls for intra-logistical purposes in warehousing. Lightweight drones are being equipped with sensor technology for use in warehouses.

Drones can be steered automatically using sophisticated software and without the need for direct human supervision. For example, car manufacturer Audi and parts maker ZF are currently using drones to carry spare parts within their manufacturing plants.

Robots in warehouses

AI robots are used to track, locate and move inventory within warehouses. Automated and intelligent robots are fully capable of moving, lifting and sorting items. Common AI robots used in warehouses include picking, forklifts and inventory.

Autonomous picking robots. Use of autonomous mobile picking robots eliminates unnecessary walking and improves efficiency of the process. Such robots typically carry carts and can be programmed to travel flexible routes in the warehouse to move products between workers and stations.

Autonomous forklift robots. Forklift robots are becoming increasingly complex and intelligent with full autonomy for some applications. These are well-suited for repetitive load-handling processes and operations. These have automated driving systems with cameras and radar systems to analyse the surrounding environment.

Autonomous inventory robots. These robots conduct their own inventory autonomously at schedules determined by the warehouse.

As per a report from *ResearchAndMarkets.com*, the global drone logistics and transportation market is anticipated to register a CAGR of over twenty per cent during the forecast period of 2019 to 2024. Freight drones are anticipated to experience the highest growth. Asia-Pacific region is expected to have the highest growth in the market during the forecast period. *JD.com*, a Chinese e-commerce company headquartered in Beijing, started making commercial drone deliveries in four select regions scattered across China since June 2016.

Logistics in healthcare

Hospitals are also using a fair amount of automation. For instance, delivery robots can move around hallways delivering pills, bringing lunch to patients, and ferrying specimen and medical equipment to different labs. Some hospitals employ robotic nurses to deal with the surge of patients and shortage of human nurses.

China is employing such robots in Wuhan hospitals during the ongoing coronavirus (COVID-19) outbreak, which has claimed thousands of lives in China and across the world. Robots are being deployed at front lines to contain the spread of the novel coronavirus. These are used for disinfection, street patrolling, and food and medicine delivery in quarantine wards.

Companion robots provide companionship to old and sick people as well as help around the house with various chores. These robots are capable of delivering food, picking up objects from around the home, finding lost things and more.

Aethon robot, called TUG, can lift a load weighing up to 200kg and autonomously drive down hospital hallways, through automatic doors, and up and down elevators. The robot is used to deliver food, medication and supplies throughout different parts of hospitals and other healthcare facilities on either a scheduled basis or on demand, at the request of a healthcare worker logged on to the system. It is designed to increase efficiency of healthcare operations.

Conclusion

Logistics is a vital part of every business model. It forms an important part of the supply chain that involves planning, implementation, forwarding and reversing the flow of goods, services and related information from the origin to the recipient. AI is transforming the logistics industry including autonomous vehicles, smart roads, automated warehousing and back-office operations. It is revolutionising logistics and supply chain management in a big way. EFY